Pharm Module 9 · Licit & Illicit Drugs & Toxicology

One-page summary · Introduction to Pharmacology (722.544) · self-directed session Wed 21 Oct 2026 · final Thu 5 Nov

It's in the exam. Joan's week 11 announcement: "*Note key words and definitions eg addiction, dependence, as the L/O for this session will be tested in the Exam.*" Short answer = **a definition + an example**.

Pusheen rule (slide 3): a slide with the cat = tested on the **concept**, not the statistics or finer details (P).

Her correction (20 Oct, mini-exam 1 Q4): **dependence = withdrawal symptoms if a prescribed drug is stopped abruptly** (B). Compulsive use with craving, loss of control and use despite harm is **addiction** (her document said C).

THE SEVEN TERMS

Term	Definition	Example
Licit	Lawful and permitted: prescription, restricted, pharmacy-only, general sale medicines + **some Class B & C** controlled drugs	Alcohol; an opioid taken as prescribed
Illicit	Unlawful / not permitted: **all Class A**, some Class B, a few Class C	Methamphetamine (Class A)
Misuse	Any substance used for a **purpose different from legal or medical guidelines** (WHO). NZ law: illegal, harmful or unauthorised use of a **controlled drug** (Misuse of Drugs Act 1975)	Taking someone else's prescription
Abuse	Self-administered in **chronically excessive amounts**, outside approved medical use or social norms, causing **physical or psychological harm**	Heavy daily drinking that damages health
Dependence	Pursued compulsively, no medical need, despite adverse effects. **Psychological**: out-of-control cravings, often denial. **Physical**: **withdrawal** when not used	Opioid stopped suddenly → withdrawal (so wean). Insulin, daily inhaler = medical dependence, **not addiction**
Addiction	A **behavioural pattern**: overwhelming involvement in **getting and using**; impairs motivation-reward, mood, behavioural inhibition; **high relapse**	Keeps using despite losing job; also gambling, eating, shopping
Tolerance	Repeated doses → **decreasing effect**, or dose must rise for the same effect (receptor downregulation)	Morphine: rapid. Marijuana: little

WITHDRAWAL & SUBSTANCE USE DISORDER (P)

Withdrawal: mild → severe, days–weeks; depends on the substance, how long, age/health, **weaned vs "cold turkey"**. E.g. insomnia, fatigue, irritability, cravings; severe: paranoia, confusion, tremors.

Substance use disorder: diagnosed with the **DSM-5**; severity = **number of criteria met** (11 criteria, incl. cravings, tolerance, withdrawal). Mild 2–3 · moderate 4–5 · severe 6+.

RISK FACTORS (YOUNG ADULTS) · WHO USES

F-I-S-B: **Family** (relatives using, home stress, childhood trauma) · **Individual** (impulsive, mental ill health, access e.g. Rx after injury) · **Social & environmental** (peers, societal acceptance, laws) · **Behavioural** (frequent use, starting "socially"). **Anyone can take drugs: avoid stereotyping.**

NZ picture (P): most adults drink, ~1 in 6 hazardously; daily vaping now above smoking; cannabis the most-used illicit drug; MDMA, cocaine, opioids rising (NZ Health Survey 2024/25, Drug Foundation 2025, Youth19).

STIMULANT OR DEPRESSANT? (P)

Stimulants ("uppers") speed up the CNS: MDMA, methamphetamine. **Depressants** ("downers") slow it: alcohol, GHB, sedatives (lorazepam, zopiclone), nitrous oxide, inhalants. **Both**: nicotine, marijuana, psychedelics, synthetic cannabinoids. Slang: P = meth · 420 = marijuana · lean = codeine + fizzy drink · pingers = MDMA · fishies = GHB · nangs = nitrous · acid = LSD.

INTERACTIONS

Medication–drug: a licit/illicit substance changes a medicine's effect → check the **NZ Formulary Interactions checker** (it lists alcohol, marijuana, nicotine, sedatives, opioids). E.g. **alcohol + another depressant** (opioid, benzodiazepine) → overdose risk; 35% of NZ fatal overdoses involved alcohol.

Drug–drug (P): Know Your Stuff combination chart, a harm-reduction chance. Caution: alcohol + nitrous; caffeine + amphetamines. Dangerous: alcohol + ketamine; opioids + cocaine; SSRI + tramadol.

NURSE'S ROLE · REFERRAL · HARM REDUCTION

Ask about use · **brief intervention** · **offer referrals** · **follow up** · **advocate**. Consider mental health services too.

Services: Community Alcohol & Drug Services (**CADS**) · **Te Ātea Marinō** (Māori) · **Tupu** (Pasifika) · **Altered High** (youth) · rehab e.g. **Odyssey House**.

Websites: DrugFoundation.org.nz · TheLevel.org.nz · AlcoholDrugHelp.org.

Harm reduction: make use safer for people who'll use anyway: legal, free **drug checking** (Know Your Stuff NZ; over half won't take a drug that isn't what they thought), **naloxone** for people at risk of opioid overdose.

TOXICOLOGY: POISONING & OVERDOSE

Poisoning: ingestion or contact (heavy metals, pesticides, household cleaners), usually accidental. **Acute** = enters in a short time · **Chronic** = gradual build-up. Most things can poison, depending on **how much**, **age & size**, already unwell, other medicines.

Poisonous = you touch/eat it (plant, berry, mushroom). **Venomous** = it bites/stings you (bee, wasp, katipō, white-tailed spider, bluebottle).

National Poisons Centre 0800 764 766 (0800 POISON) · free · 24 h. Swallowed medicine/product/chemical/plant · **overdose** · inhaled fumes · splash on eyes/skin · symptoms from recreational drugs. **Unconscious or not breathing → 111.**

ANTIDOTES

Counteract a poison by: **binding** it (activated charcoal) · **enzyme inhibition** · **enzyme activation** · **blocking receptors** (antagonists).

Naloxone (Narcan): **opioid antagonist**, reverses opioid overdose. **N-acetylcysteine (NAC)**: reduces the **toxic metabolites of paracetamol**.

HER FORMATIVE QUESTIONS

- Mini-exam 1 Q4: dependence = **withdrawal if a prescribed drug is stopped abruptly** (B, corrected in class).
- Test three Q5: addiction = **a behavioural pattern … overwhelming involvement in getting and using the drug**.
- Mini-exam 2 Q11: controlled drugs = **under special legal controls to reduce misuse and abuse** (not "illicit").
- Mini-exam 3 Q4: a half dose by mistake is a medication error. Calling 0800 POISON is **not** appropriate.

Controlled-drug classes (Legal module): **A** very high risk, never prescribed (heroin, meth) · **B** high risk, prescribed, double-locked (morphine) · **C** lower risk (codeine).

(P) = a Pusheen slide: know the concept, not the numbers or lists. Sources: Licit and Illicit Drugs deck (slides 2–40 + speaker notes; Knights et al. 2023, WHO 1994, APA 2022, Alhammad et al. 2022, Te Whatu Ora); Joan's announcement and 20 Oct revision session; Formative mini-exams 1–3, test three; For-interest articles (NZ Herald 2023–24, Stuff 2022). Study aid, not clinical guidance. jeremyspm.github.io/pharm-final